

Simon's Sequence

A Memory Tile Game Built with 2D Computer Graphics Algorithms and Digital Image Processing Techniques

Game Instructions and Rules | CSE 452 – Graphics and Image Processing

1. Game Overview

Simon's Sequence is a memory game in the style of the classic "Simon Says". The board shows a set of colored tiles. The game flashes the tiles in a random order, and the player must click them back in exactly the same order. Each time the player succeeds, one new tile is appended to the sequence, so the sequence grows longer and harder to remember every round. There is no fixed "win" condition — the sequence keeps growing indefinitely. The round only ends when the player runs out of lives, so the objective is to reach the highest possible score.

2. Starting the Game

- Launch the game. The **Main Menu** appears first.
- Choose a **Difficulty**: Easy, Normal, or Hard.
- Choose a **Board Size**: 4 tiles or 6 tiles.
- Click **Start Game** to begin.

3. How a Round Works

Each round follows the same repeating cycle:

- 1 **Step 1 – Watch:** The game flashes the tiles in the current sequence, one at a time, in order.
- 2 **Step 2 – Repeat:** Click the tiles in the exact same order you just saw, before the countdown timer runs out.
- 3 **Step 3 – Grow:** If the whole sequence is matched correctly, one new random tile is appended to the sequence and the next round begins — the sequence is now one tile longer.
- 4 **Step 4 – Mistake:** Clicking the wrong tile, or letting the timer reach zero, costs one life and restarts the current sequence from the beginning (same length, watch it again).

Because tiles are chosen randomly, colors can repeat within a sequence — for example on a 4-tile board the sequence can easily grow past 4 steps, reusing colors in a new order each time.

4. Scoring, Combo, and Lives

Every correct tile click adds points to the score. Consecutive correct clicks build a **combo**, which increases the points earned per click the longer the streak continues. A wrong click or a timeout resets the

combo to zero and removes one life. The number of starting lives depends on the chosen difficulty (see the table below). When lives reach zero, the game ends and, if the final score beats the saved **High Score**, it is stored automatically for next time.

5. Difficulty Levels

Difficulty is fixed for the whole game once selected in the menu — it controls how fast tiles flash, how much time is given to answer, and how many lives are available. It does not increase automatically as the sequence grows; the growing sequence itself is what makes later rounds harder.

Difficulty	Flash Time	Gap Between Flashes	Time to Answer	Lives
Easy	0.65 s	0.30 s	6.0 s	5
Normal	0.45 s	0.18 s	4.0 s	3
Hard	0.30 s	0.10 s	2.5 s	1

Lower flash time and gap mean tiles light up faster and are harder to track visually. Lower answer time means less time to click the sequence back. Fewer lives mean fewer mistakes are allowed — Hard mode allows only a single mistake.

6. Tile Colors and Live Filter Effects

Every tile color is linked to a specific image-processing filter. Each time that tile is clicked correctly as part of the sequence, its filter is applied live to the game screen for a brief moment, turning every correct click into a visual demonstration of that filter.

Tile Color	Filter Applied	Board
Blue	Mean / Blur filter	4-tile and 6-tile
Red	Sharpen filter	4-tile and 6-tile
Green	Sobel edge detection	4-tile and 6-tile
Yellow	Emboss filter	4-tile and 6-tile
Purple	Median filter	6-tile only
Cyan	Sharpen + edge detection combo	6-tile only

7. Controls

Input	Action
Left mouse click	Select a tile / click a menu button
P	Pause or resume the game
Esc	Quit the game

M (on Game Over screen)	Open a side-by-side comparison of the final screen with mean, sharpen, Sobel ed
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8. Game Over

The game ends when all lives are lost. The final score is shown along with the current High Score; if the new score is higher, it is saved automatically. Clicking anywhere returns to the Main Menu to start a new game.

9. Quick Tips

- Start on Easy to learn the tile layout before trying Normal or Hard.
- Watch the countdown bar at the top — it shows how much time remains to answer.
- Keep the combo alive for extra points — every mistake resets it to zero.
- On Hard mode, a single mistake ends the game — play cautiously.